HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MASTER'S THESIS

Privacy-First Classroom Engagement Analysis System via Edge Computing and Lightweight Deep Learning Models

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THESIS TOPIC

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Acknowledgements

[To write by hand: acknowledgements, 100–150 words.]

Abstract

Camera-based attention monitoring in schools has drawn strong objections where it recognised students' faces, sent footage to servers and scored individual students. This thesis asks whether the useful part, a class-level indicator of how many students appear distracted, can be built without that surveillance.

The system, hcpose, runs entirely on a Raspberry Pi 5 with a Hailo-8 accelerator. A YOLOv8-pose model reduced to 13 upper-body keypoints finds the students. No face is recognised, frames stay in memory, and students exist only as anonymous seats. Per-seat calibration, two head-turn measures and smoothed rules decide whether each seat looks focused. The teacher sees only the share of the class that appears distracted, with a 95% Wilson interval.

On the device, the deployment setting (the n model, with tiling of the far rows) runs in real time at 30 FPS with 45 ms latency. It draws about 4.3 W at the board plus about 2.5 W for the accelerator, without thermal throttling. Tiling raises the share of students found on the SCB dataset from 49% to 76%, and from 17% to 74% in the far third, without retraining. On COCO val2017, the deployed INT8 models lose 5-7 keypoint AP points against their PyTorch versions.

Classifiers trained on SCB do not transfer to real lessons: they flag 48-89% of an attentive lecture as distracted. The calibrated rules keep false alarms low, with 90-91% specificity on 451 human labels from three clips, including a 10-student class recorded after every threshold was fixed. With a nose-to-shoulder head-turn measure that keeps working in profile, they find 45-56% of scripted distraction. Phone use remains invisible to pose. The results show that an aggregate, on-device indicator is feasible, while reliable per-student judgement needs classroom-specific training labels.

Keywords: student engagement, pose estimation, edge computing, privacy by design, Hailo-8, Raspberry Pi

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CHAPTER 1. INTRODUCTION

1.1 Motivation

A teacher facing 30-50 students cannot see every student's attention at every moment, and a simple class-level indicator of how many students seem distracted could help them pace a lesson. Camera-based systems that do this already exist, but the best known of them, in Chinese schools in 2018-2019, recognised every student's face, sent footage to servers, and scored individual students for teachers and parents. They drew public backlash and one was suspended by the authorities (chapter 2).

The question this thesis asks is whether the useful part, an aggregate attention indicator, can be built without the surveillance. Three design choices follow from that:

- **Pose instead of faces.** Only body keypoints (13 upper-body points) are extracted; no face is recognised or stored, and students are only anonymous seat numbers within one session.
- **Edge instead of cloud.** Everything runs on a Raspberry Pi 5 with a Hailo-8 accelerator in the room. No frame leaves the device, and by default none is stored.
- **Class level instead of individual scores.** The output meant for the teacher is the share of the class that appears distracted, with an honest uncertainty range, not a per-student report.

The remaining problems are technical: lightweight pose models miss small students in the back rows, and "looking away" is hard to judge from a skeleton across different classrooms and camera positions.

1.2 Research questions

1. **RQ1, feasibility at the edge.** Can a pose-based engagement pipeline run in real time on a Raspberry Pi 5 with a Hailo-8 for a full classroom, and at what power and thermal cost?
2. **RQ2, coverage.** How many students does a lightweight pose model find in a real classroom, and can pre- and post-processing raise that without retraining the model?
3. **RQ3, the per-student decision.** How well do pose-based decisions, a calibrated rule set and classifiers trained on a public classroom dataset, separate focused from distracted students in real classrooms?
4. **RQ4, the class indicator.** Can a class-level share with a trustworthy uncertainty range be built when many students are missed?

1.3 Contributions

- **hcpose**, an open-source runtime for custom YOLOv8-pose models on the Hailo-8, with a GStreamer engine and a Python engine that give the same results, installable and runnable with one command.
- **Tiling for far rows:** extra inferences on zoomed tiles of the back of the room, merged without duplicates in C++ inside the GStreamer pipeline. On SCB images it raised the share of students found from 49% to 76%, and from 17% to 74% in the far third, without changing the model.
- **Duplicate-skeleton removal:** a post-processing step that removes the second box YOLO-pose often gives one student.
- **An engagement analyzer** with per-seat calibration and two head-turn measures. The second measure, the nose's offset from the shoulder centre, keeps working in profile, where the usual ear-based estimate fails.
- **A class estimate** that remembers seats, weights near, middle and far zones, and reports a Wilson interval, with seat counting that resists spurious detections.
- **An evaluation on four kinds of evidence:**
  - the SCB dataset
  - four real classroom clips (three public, one recorded for this work)
  - 451 blind human labels from three clips
  - a scripted staged session

  Together they show that classifiers trained on SCB do not transfer to real lectures, while per-student calibration does.
- **Edge measurements** on the device: throughput, latency, CPU, temperature and board power.
- **A privacy analysis** against Vietnam's Personal Data Protection Law (2025) and Decree 356/2025.

1.4 Scope

The system estimates visible attention from posture and head direction; it does not measure learning, emotion or cognitive engagement, and it cannot see a phone held like a pen. The pose models were trained before this work and kept fixed: every improvement is in the runtime, the post-processing and the analysis. The evaluation uses short recordings and staged sessions rather than a long field deployment.

1.5 Structure

Chapter 2 reviews engagement detection, pose estimation at the edge and the deployed systems that raised privacy concerns. Chapter 3 describes the system. Chapter 4 evaluates it and analyses its privacy properties and legal fit. Chapter 5 answers the research questions and lists future work.

CHAPTER 2. RELATED WORK

2.1 Approaches to classroom engagement detection

Automatic engagement detection in classrooms has used four kinds of signal: faces, body pose, body orientation and wearable sensors. They trade accuracy against intrusiveness.

Faces. Facial expressions and gaze carry rich information, but they need a frontal, well-resolved face for each student. That fails in large classes, where back rows are small and often occluded [1]. Face analysis also identifies the student, which is the root of the privacy problems in section 2.3.

Body pose. Skeleton keypoints are less identifying and survive low resolution better.

- Tarigan et al. [2] detect students with YOLOv8m, extract seven keypoints per person with MediaPipe, and classify them with a small fully connected network into engaged, bored, raising a hand and on the phone.
  - **Data:** about 10,000 frames recorded with a laptop webcam in a small lab setting.
  - **Result:** 0.70 accuracy, against 0.41 for a YOLOv4 and OpenPose baseline.
  - **Weaknesses they report:** phone use was mostly confused with bored or engaged, and overlapping boxes caused double detections.
  - **Gaps:** privacy and deployment hardware are not discussed.
- Public datasets such as SCB [3] label classroom actions (reading, writing, raising a hand, turning the head, discussing) rather than attention, and are the usual training source.

Body orientation. Chen et al. [1] estimate each student's body orientation jointly with detection, using a one-stage YOLOv11-based model (JointBDOE). They read engagement from how often orientation changes and whether it follows the teacher.

- **Orientation accuracy:** 10.6° mean error on MEBOW.
- **Real classrooms:** ten lessons from kindergarten to university, with faces blurred.
- **Gaps:** no per-student engagement labels and no quantitative engagement accuracy. The models were trained on six RTX 4090 GPUs, and inference hardware is not stated.

Wearables. EEG headbands such as BrainCo's FocusEDU measure brain activity directly, but they require every student to wear a device and drew strong objections (section 2.3).

Review. The same group's systematic review [4] followed PRISMA 2020. It searched Scopus for 2019–2023 and narrowed 547 papers to 12 studies. Its main findings:

- **Methods:** YOLO detectors were the most used (5 of 12 studies), then VGG classifiers (4). Hybrids that split detection from pose did best among skeleton methods: YOLOv4 with OpenPose reached 86.1% accuracy on four classes.
- **Pose alone:** the pose-only study scored lowest. It found phone use with precision 0.22 and recall 0.11, the behaviour that the 13 keypoints here also cannot separate from writing (section 4).
- **Challenges:** costly manual annotation, class imbalance, small and occluded students in crowded rooms, and compute cost. The review points to lightweight YOLOv8 models and notes that keypoints help preserve privacy.
- **Comparability:** the studies use different classes and metrics, so their results cannot be ranked directly.

This thesis takes up the review's two open directions: a lightweight YOLOv8-pose model on edge hardware, and keypoints as a privacy measure.

Two gaps run through this work:

- Accuracy is reported on lab or action datasets rather than on attention in real lessons.
- Privacy and the computing platform are rarely part of the design.

2.2 Pose estimation on edge hardware

Neural accelerators have made multi-person pose estimation feasible on small boards. The Hailo-8 delivers 26 TOPS at a typical 2.5 W [5]. As the Raspberry Pi AI HAT+, it attaches to a Raspberry Pi 5 over PCIe. Hailo's example applications run a YOLOv8-pose model in a GStreamer pipeline and hand each frame's detections to a Python callback.

Core Electronics' guide [6] builds on that pipeline to read 17 COCO keypoints and drive single-person demos, such as gesture control and a game. It reports no frame rates. It also notes practical limits:

- some capture resolutions caused crashes
- a start-up delay
- extended use may need cooling

These examples target one stock COCO model and one or a few people near the camera. A classroom differs in four ways:

- a custom model (13 upper-body keypoints)
- 20-50 people at very different scales
- a camera far from the back rows
- the need to keep the full pipeline, including analysis, on the device

Section 3 describes the runtime built for that case.

2.3 Deployed systems and the controversies they raised

The best-documented deployments of attention monitoring in schools are from China in 2017-2019. All three below were criticised for exactly the properties this thesis avoids: identifying students, scoring them individually, and doing so without consent.

Hangzhou No. 11 High School, 2018. [7–10]

- **Vendor:** Hikvision's "smart classroom behavioural management system".
- **How it worked:** cameras above the blackboard scanned the class about every 30 seconds. They recognised each student and classified six behaviours (reading, writing, listening, standing, raising a hand, lying on the desk) and seven facial expressions.
- **Who saw the scores:** they appeared on classroom screens, with attention rankings in the hallway.
- **Data:** the school said data stayed on a local server and no images were kept.
- **Response:** educators objected to the lack of consent and questioned whether expressions say anything about learning. After the media coverage in May 2018, the school stopped scoring facial expressions.

Hanwang Education's Class Care System, 2017-2019. [10]

- **How it worked:** a dome camera photographed the class once per second. The footage went to a server, where deep learning identified each face and sorted behaviour into listening, answering, writing, interacting and sleeping.
- **Who saw the scores:** each student received a weekly score in an app for teachers and parents.
- **Rollout:** piloted in seven schools without Ministry of Education approval.
- **Students' response:** at one school students were not told about it. The company's own manager said students "hate it", and students at another school unplugged the cameras before exams.

BrainCo FocusEDU headbands, Jinhua, 2019. [11]

- **How it worked:** a primary school had pupils wear brainwave-sensing headbands during training sessions. According to the vendor, teachers saw only the class average.
- **Response:** after a related Weibo topic passed 220 million views, Zhejiang education officials ordered the school to suspend their use.

Two lessons carry into this work:

- Identity and individual scores are what made these systems objectionable, more than the camera itself.
- Centralising footage on servers widens who can see it and what it can be reused for.

At that time China had no specific law on facial or biometric data. Vietnam now does, through its Personal Data Protection Law (2025) [12], discussed in section 4.

2.4 The gap this thesis addresses

No prior system combines the four properties this work targets:

- sensing without identity
- processing entirely on a device in the room
- an aggregate rather than per-student output
- an evaluation of attention in real classrooms, including on the deployed hardware

Table 2.1 Prior systems and studies compared with this thesis

Work	**Sensing**	**Where it runs**	**Output for the teacher**	**Identifies students**	**Evaluated on real lessons**	**Measured on edge hardware**
Hangzhou No. 11 / Hikvision (2018)	faces, behaviour	local server	per-student scores, rankings	yes	not published	no
Hanwang Class Care System	faces, behaviour	server	weekly per-student score	yes	not published	no
BrainCo FocusEDU (2019)	EEG headband	device + app	class average (vendor)	per device	not published	–
Tarigan et al. [2]	7 keypoints	not stated	per-person class	no	no (lab webcam)	no
Chen et al. [1]	body orientation	GPU (not stated)	orientation patterns	no	qualitative only	no
Core Electronics guide [6]	17 keypoints	Pi 5 + Hailo-8	demo only	no	no	partly
This thesis	13 upper-body keypoints	Pi 5 + Hailo-8, offline	class share + 95% range	no (anonymous seats)	yes: clips, human labels, staged sessions	yes: FPS, latency, temperature, power

CHAPTER 3. SYSTEM DESIGN AND IMPLEMENTATION

3.1 Architecture

The system has two stages on one device. The detection runtime turns camera frames into skeletons on the Hailo-8, and the engagement analysis turns skeletons into a class-level share. Frames exist only in memory, and per-student states are never output by default.

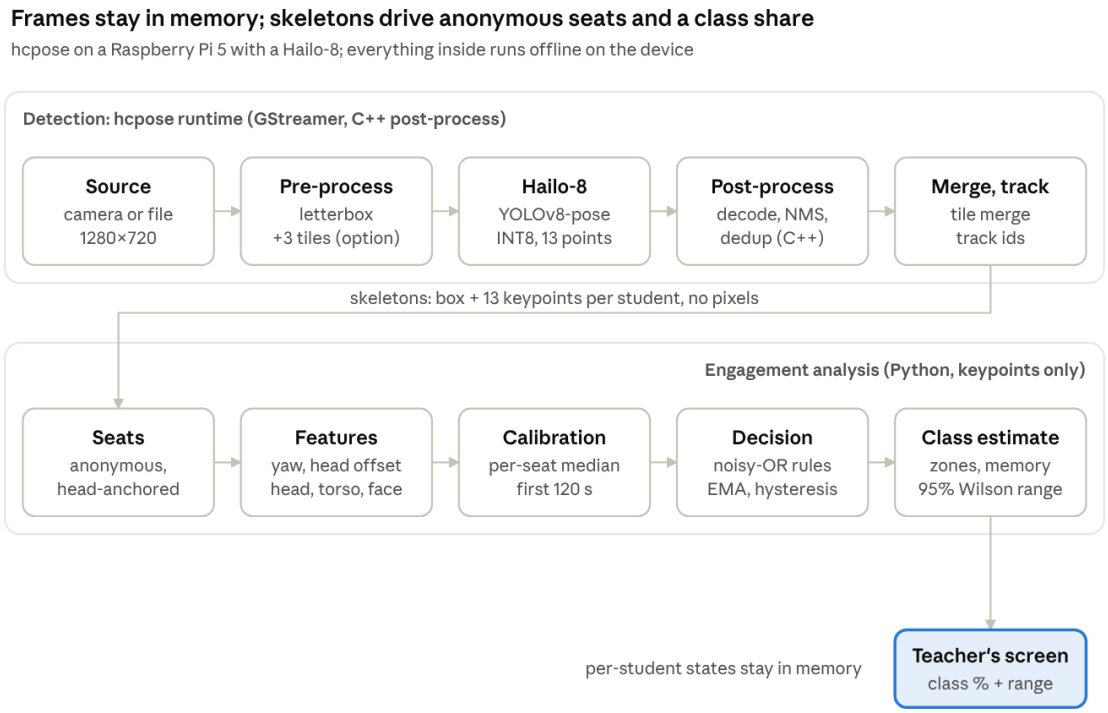


Figure 3.1 Architecture of hcpose: frames stay in memory, and skeletons drive anonymous seats and a class-level share

The skeleton file (`--save-json`) and an annotated video (`--output`) can be written at the hand-off between the two stages for development and evaluation. Both are off by default. The data-flow view used for the privacy analysis is in chapter 4.

3.2 Pose models and compilation

The pose models are YOLOv8-pose networks in three sizes, n, s and m, that predict one class (person) and 13 upper-body keypoints: COCO keypoints 0-12 [13], from the nose to the hips. Classroom desks hide the legs, so the lower-body points carry no information. The models were trained on COCO with Ultralytics [14], in the companion repository `hailo-custom-yolo-pose`, and kept fixed for this thesis.

Training. All three sizes start from Ultralytics' COCO-pretrained YOLOv8-pose weights. They were fine-tuned on COCO 2017 person keypoints: train2017 for training, val2017 for validation. The labels were converted with Ultralytics' `convert_coco`

Figure to add: The device

put the image at `figures/manual/device.png` (or .jpg/.pdf)

Figure 3.2 The device: Raspberry Pi 5 with the Hailo-8 AI HAT+ (26 TOPS), active cooler and camera.

and cut from 17 to 13 keypoints. The settings were the same for every size except the batch:

- 100 epochs at 640×640; batch 16 for n and m, 32 for s; seed 0, deterministic mode, mixed precision.
- `optimizer=auto`: Ultralytics chose the optimizer and learning rate itself, so the script's `lr0=0.01` and `momentum=0.937` were not used. The logged rates show a 3-epoch warm-up, then a linear fall to 1% of the peak by epoch 100.
- Loss weights and augmentation at Ultralytics defaults: HSV jitter, scale 0.5, translate 0.1, and horizontal flip at 0.5 with left and right keypoints swapped. Mosaic was on except in the last 10 epochs.

Training ran on the project's workstation (one NVIDIA RTX 3060) with Ultralytics' June 2026 release.

Table 3.1 Training of the three pose models

Model	Best epoch	Training time (h)	Box mAP50-95 (training log)
n	100	43.1	0.527
s	99	42.4	0.584
m	96	89.9	0.619

Keypoint accuracy. On COCO val2017 the trained models reach 0.470, 0.568 and 0.625 keypoint AP on their 13 points, 4-5 points below the stock 17-keypoint models on the same points. On the Hailo-8, INT8 costs n 5.3 points and m 6.8. The s HEF loses 30.8 points, so it is left out of every accuracy result.

Keypoint AP (OKS 0.50:0.95) on the 5,000 val2017 images, scored with pycocotools [15] and COCO's per-keypoint tolerances. Every model gets the same 640×640 letterbox. As a check on the scoring, the stock models score 0.499, 0.594 and 0.645 on all 17 points, within 0.6 points of Ultralytics' published 50.4, 60.0 and 65.0.

- **Size.** m gains 0.14 AP over n on the Hailo-8 (0.557 against 0.417), for 4.7×

Table 3.2 Keypoint AP on COCO val2017, 13 upper-body keypoints, COCO tolerances

Model	Stock 17-point model, on the 13 points	Trained 13-point model (.pt)	INT8 HEF on the Hailo-8	INT8 cost (AP points)
n	0.523	0.470	0.417	−5.3
s	0.613	0.568	0.260	−30.8
m	0.665	0.625	0.557	−6.8

the latency.

- **Training on 13 points.** Most of the 4-5 point loss is in placing the face points precisely. With one loose tolerance for every point, the gap shrinks to 1.1-1.5 points.
- **The s HEF.** It still finds people (AP50 0.683) but places keypoints poorly (AP75 0.127). On matched detections its median keypoint error is 4-9% of the box size, against 2-3% for the n and m HEFs, and every point is shifted the same way. The n model was compiled with the same script and is unaffected, so the fault is in this compile, not in the settings. The s model appears only in the speed results.
- **Training-log scores.** The log's keypoint mAP50-95 (0.329, 0.376, 0.401) is far below the same models scored by pycocotools with the same uniform tolerance (0.579, 0.660, 0.706). The cause is people with no labelled keypoint among the 13. They make up 4,503 of the 10,777 people in the validation labels (41.8%). pycocotools ignores them, but Ultralytics' validator counts them as targets it can never match. With those people removed, Ultralytics' own validator gives 0.555, 0.632 and 0.669, close to the pycocotools figures. The log therefore understates keypoint accuracy, and its keypoint scores are not used as results.

For the Hailo-8 each model is quantised to INT8 and compiled with the Hailo Dataflow Compiler into a HEF:

- **No on-chip NMS.** The nine raw head outputs are exported as end nodes: box distribution (DFL), class score and keypoints at strides 8, 16 and 32. Decoding and NMS run on the host, which keeps the HEF independent of the model's keypoint count.
- **Sigmoid inside the HEF.** The class scores are passed through sigmoid inside the HEF. The runtime detects this per model, which fixed a double sigmoid in the code it replaced.

Quantisation. The compiler (Hailo Dataflow Compiler 3.33.1, Model Zoo 2.18) calibrated the INT8 ranges on 1,024 random train2017 images resized to 640×640. The n and s models share one model script and m has its own. Both scripts keep three of the nine output layers at 16 bits; by their order these are the keypoint outputs. Both run Hailo's post-quantisation fine-tuning at learning rate 0.00015, and

Table 3.3 Throughput and latency of the INT8 HEFs on the Hailo-8 alone

Model	Hailo-8 FPS	Hardware latency (ms)
n	131.5	5.7
s	69.5	11.9
m	31.5	26.7

the n/s script also turns off pre-quantisation equalisation.

3.3 Runtime: hcpose

hcpose runs any YOLOv8-pose HEF on a Raspberry Pi 5. It reads the keypoint count, strides, tensor order, output data type and score activation from the HEF at start-up, so the same command runs the 13-keypoint models, the stock 17-keypoint model or others.

Two engines, one result.

- **GStreamer engine** (default, for live use): the camera or file source feeds `hailonet`. A C++ `hailofilter` decodes and post-processes the outputs, `hailotracker` assigns track ids, and `hailooverlay` draws.
- **Python engine** (for images, debugging and evaluation): OpenCV and HailoRT with a NumPy decoder.
- **Equivalence:** the two decoders produce identical detections, checked by the tests on recorded model outputs.

Post-processing.

- **Decoding:** DFL boxes and keypoints are decoded and mapped back through the letterbox.
- **NMS:** candidates are sorted by score first. The inherited code skipped this, so overlaps were resolved by anchor order.
- **Duplicate-skeleton removal:** after NMS, a box is dropped when it lies inside a stronger box and its skeleton matches: at least three common keypoints, closer on average than $0.1 \times \sqrt{}$(smaller box area). YOLO-pose often outputs a tight and a loose box for one student, and box NMS keeps both.

Tiling for far rows. With `--tiles`, the model also runs on a row of overlapping, roughly square tiles over the top 60% of the frame (three tiles at 1280×720).

- **In the GStreamer engine:** a custom crop function hands `hailocropper` the full frame plus the tiles. A second `hailofilter` after the aggregator merges the results.
- **The merge:** tile detections are mapped to frame coordinates, those cut by an inner tile edge are dropped, and duplicates across passes are merged by IoU, or by containment plus agreeing keypoints.
- **Checks:** the C++ merge equals the Python merge on 25 random scenes.

- **Cost:** tiling multiplies inference by four.

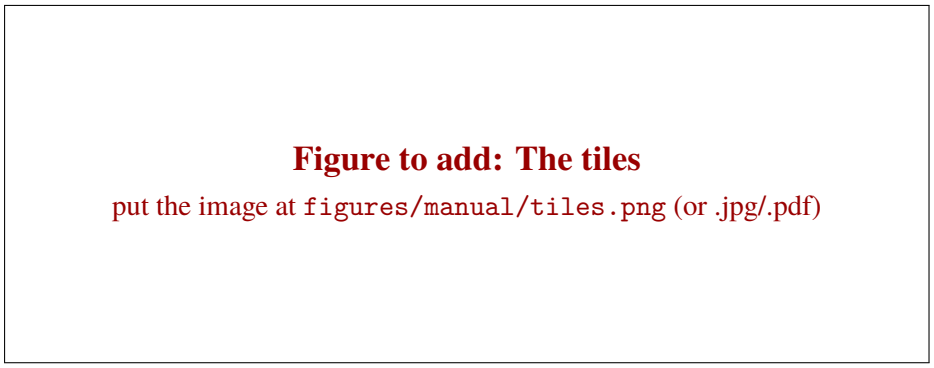


Figure 3.3 Detections on the exam-hall clip without (left) and with (right) the three tiles over the far rows.

Outputs.

- The annotated window, or headless operation.
- Optionally an annotated video (`--output`), for development only.
- Optionally a JSON-lines skeleton file (`--save-json`) with boxes and keypoints, no pixels.

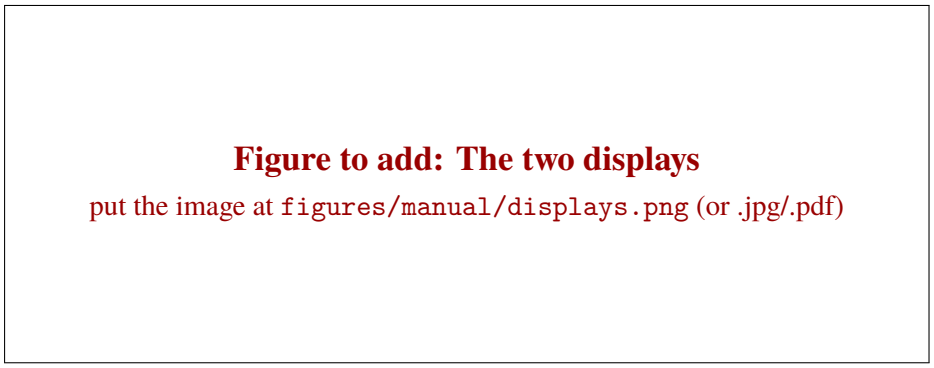


Figure 3.4 Left: development view with per-student states (green focused, red distracted, white calibrating). Right: the teacher's view, with the class share and its 95% range only.

3.4 Engagement analysis

The analyzer turns each frame's skeletons into a focused, distracted or unknown state per student and a class-level share with a range. It works on keypoints alone and can replay a saved skeleton file (`hcpose analyze`) with any settings.

Seats. Students are tracked as anonymous seats, matched frame to frame by the top centre of their box (the head). The head stays put when a student leans or turns around, while the box centre jumps. A seat counts toward the class only while it is current and consistently seen.

Features, normalised by shoulder width:

- **Two head-turn measures.** Ear-based yaw: the nose between the ears. Head offset: the nose relative to the shoulder centre, which keeps working in profile.

- **Head height** above the shoulder line.
- **Torso ratio**, which shrinks when the body rotates.
- **Face hidden** while the shoulders are visible.
- **Hand raised**, and **hand at face**.

Per-seat calibration.

- Each seat learns its own attentive pose as the median of its first 120 s of usable observation. Before that, partial medians and the room's median are used.
- The seat's yaw and head-offset baselines are kept within 0.3 and 0.15 of the room's, so a student who looks away during calibration does not learn "turned" as normal.
- `--camera-position` (left, centre, right) sets a prior for the room's attentive direction.

Decision. Each cue is a ramp from "normal" to "clearly off" relative to the seat's baseline. They combine with a noisy-OR, so weak cues such as head down (writing looks the same) only count together with another cue:

$$s = 1 - \prod_{c} (1 - w_c \, r_c) \tag{3.1}$$

Here r_c is cue c's ramp value and w_c its weight. A raised hand overrides the score to 0. The default weights are: head turned 1.0, face hidden 0.9, head down 0.7, body turned 0.5, hand at face 0.4. The learned classifiers (trees, logistic regression, MLP) can replace the noisy-OR, but are off by default (chapter 4).

Smoothing. The score is smoothed with an exponential moving average (τ = 1 s) and switched with hysteresis:

- **to distracted:** above 0.6 for 2.5 s
- **back to focused:** below 0.4 for 1.5 s
- **unknown:** no usable keypoints for 2 s

This keeps brief glances from flipping a student.

Class estimate.

- Seats keep their last state for 15 s after they were last seen.
- They are grouped into near, middle and far zones by image height, and each zone is weighted by its number of seats.
- The class share comes with a 95% Wilson interval [16] around the shown (smoothed) value. It is narrowed by the finite-population correction only when the real head count is given (`--class-size`).
- On screen: "43% distracted (20-65%) | 14 of ~30 students".

3.5 Configuration, testing and reproducibility

Configuration. Every threshold, weight and option is a setting in `configs/engagement.yaml` or a command-line flag:

- camera position
- display mode (per-student, class, both or none)
- class size
- calibration file
- classifier
- tiling

A per-model sidecar YAML sets detection thresholds and keypoint names.

Tests. 44 automated tests cover:

- both decoders against recorded HEF outputs and Ultralytics' own decoding
- the C++ letterbox mapping and tile merge against Python
- the duplicate removal
- the engagement logic on synthetic classes: calibration, glances, missing frames, seat counting, the interval, the head-offset measure
- the staged-session scorer on a synthetic session with a known answer

Reproducibility. Results and the tools that made them are in the repository:

- skeleton files from the Pi
- edge logs
- the human labels
- staged-session scripts
- threshold sweeps

Every analysis can be rerun from the saved skeletons without the camera footage. The evaluated version is tagged `v1.0-thesis`.

CHAPTER 4. EVALUATION AND PRIVACY

4.1 Evaluation setup

The system is evaluated at four levels: detection coverage, the per-student decision, the class estimate, and on-device performance. Several sources of evidence are used, because no single dataset has per-student attention labels for real lectures.

System under test. hcpose on a Raspberry Pi 5 with a Hailo-8 AI HAT (26 TOPS), Raspberry Pi OS trixie, HailoRT 4.23 / TAPPAS 5.1. The pose models are the project's own YOLOv8-pose upper-body models (13 keypoints) in three sizes, n, s and m, compiled to INT8 HEFs. Model weights were kept fixed throughout: all improvements are in pre- and post-processing and in the engagement logic.

Evidence

Table 4.1 Sources of evidence used in the evaluation

Source	What it is	What it measures	Labels
SCB-Dataset (three subsets) [3]	8,116 unique classroom images after merging duplicates, Chinese K-12	detection coverage, per-student classifier accuracy (lesson-grouped 5-fold CV)	action boxes: read, write, hand-raising (focused); TurnHead, BowHead, discuss (distracted)
Exam-hall clip [17]	72 s, high corner camera, about 30 adult students writing	false alarms, class estimate	none; context implies a share near 0
Study-hall clip [18]	6 min, camera at desk height, about 50 students	dense rooms, seat counting, low-camera failure	none; context implies a low share
Lecture clip [19]	3 min 18 s, high front camera, about 35 students, mostly attentive	transfer of the classifiers to a real lecture	none; context implies a share near 0
Small-class clip (clip_5)	14 min, iPhone mounted high at the front, 10 secondary-school students in three rows; a rough script the students did not consistently follow	false alarms and catches on a class nothing was tuned on; seat counting at close range	150 student-moments labelled blind

(continued)

Source	What it is	What it measures	Labels
Staged two-person session (clip_4)	4 min 45 s, phone mounted high at the front, two adults acting 7 scripted events	recall, precision, latency, false alarms during writing	exact per second, from the script
Human labels	301 student-moments from two clips, labelled blind	precision and recall on real classes	focused / distracted / can't tell

Metrics. Coverage = share of labelled students matched to a detection (centre coverage ≥ 0.5 or IoU ≥ 0.3). Per-student: AUC, within-image AUC (only images with both classes, which removes the "recognise the image" shortcut), balanced accuracy, precision and recall of *distracted*. Class level: mean class distracted %, the width of its 95% range, and inferred class size. Device: FPS, latency, CPU, memory, temperature, throttling, power.

SCB numbers come from the deployed INT8 m HEF unless marked. Clip numbers come from the INT8 HEFs on the Pi; the two charts and the human-label scores use the same n model in FP32 on a PC, which the INT8 runs reproduce within a few points.

4.2 Detection coverage and tiling

Without tiling, the model misses most students in the far rows; running it again on three zoomed-in tiles of the top 60% of the frame lifts coverage from 48.6% to 75.5% of labelled students at threshold 0.3, and from 17.0% to 74.4% in the far third.

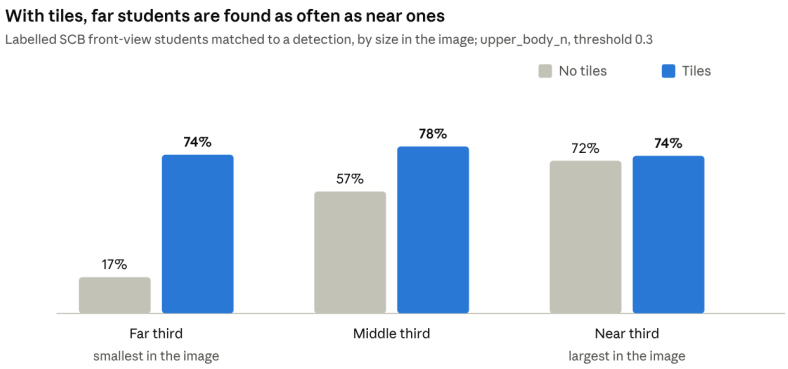


Figure 4.1 Labelled SCB front-view students detected, by size in the image, without and with tiles (upper_body_n, threshold 0.3)

Method. Besides the full frame, the model runs on one row of overlapping, roughly square tiles over the top 60% of the image (three tiles at 1280×720). Tile detections

are mapped back to the frame; those cut by an inner tile edge are dropped, and duplicates across passes are merged by IoU, or by containment plus agreeing keypoints. Both engines implement the same rules; the C++ merge in the GStreamer pipeline was checked equal to the Python merge on 25 random scenes. The detector weights are unchanged.

On the deployed INT8 m model, the number of matched SCB front-view students rose from 23,186 without tiles to 36,226 with them (+56%); 13,269 students were found only with tiles, 229 only without.

On the classroom clips, students per frame:

Table 4.2 Students detected per frame on the classroom clips, without and with tiles

Clip	Model (INT8, Pi)	No tiles	Tiles
Exam hall	n	12.6	18.6
Exam hall	m	14.2	22.1
Lecture	n	15.6	17.9
Lecture	m	20.1	22.0
Study hall	n	10.3	23.5
Study hall	m	10.6	21.0

Cost. Full frame plus three tiles is four inferences per frame, and the frame rate drops by exactly four: n from 131 to 32.8 FPS (still real time) and m from 31.5 to 7.9 FPS (enough for engagement, which needs 5-10 FPS, but not for a live camera without dropping frames). The tiles help most in dense rooms with far rows, doubling the students found in the study hall; in the lecture clip, where the far rows are already large enough, they add only two to three per frame.

4.3 Duplicate boxes and the class estimate

Two post-processing fixes made the class-level number trustworthy: removing same-skeleton duplicate boxes, and counting only current, consistently seen seats. On the device, duplicates fell from 0.9-1.6 per frame to about 0.01, and the inferred class size now tracks the real head count.

Same-skeleton duplicates. YOLO-pose often outputs a tight and a loose box for one student with the same skeleton. Their box IoU is low, so box NMS keeps both, and the analyzer then counts one student twice. After NMS, a box is dropped when it lies inside a stronger box ($\geq 70\%$ of the smaller box) and at least three keypoints visible in both are, on average, closer than $0.1 \times \sqrt{}$(smaller box area). The same rule runs in the NumPy and the C++ post-process.

The small coverage loss is mostly the SCB matcher having credited a second student with the duplicate box.

Class estimate. A plain share of *detected* students is biased, because missed students are mostly in the far rows and not missed at random. The estimate therefore:

1. remembers each seat's last state for `presence_seconds` (15 s), so a student

Table 4.3 Effect of same-skeleton duplicate removal

Measure	Before	After
SCB detections removed (1,500 images)	–	6.5% (6.3% with tiles)
Duplicate pairs per frame, exam clip, n, no tiles (Pi)	1.63	0.01
Duplicate pairs per frame, exam clip, m, tiles (Pi)	1.54	0.00
SCB coverage change	–	−0.6 to −0.8 points

Figure to add: A same-skeleton duplicate

put the image at `figures/manual/dedup.png` (or .jpg/.pdf)

Figure 4.2 One student, two boxes on the same skeleton (left); after the same-skeleton check (right).

missed for a few frames still counts;

2. splits the seats into near, middle and far zones by image height, and weights each zone by its number of seats rather than by how many students happen to be detected there;
3. counts a seat toward the class size only if it was seen in the last 60 s and, before calibration, in at least 15% of frames;
4. puts a 95% range around the shown (smoothed) estimate.

The range is a Wilson score interval [16] for a share p measured on n of N students. With a known class size N (`--class-size`), it is narrowed by the finite-population correction, shrinking toward p so p stays inside:

$$\tilde{p} = \frac{p + \frac{z^2}{2n}}{1 + \frac{z^2}{n}}, \qquad h = \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{p(1-p)}{n} + \frac{z^2}{4n^2}}, \qquad f = \sqrt{\frac{N-n}{N-1}} \tag{4.1}$$

with z = 1.96; the range is [p̃ − h, p̃ + h] clipped to [0, 1], and with a known N its ends move toward p by the factor f. Without a known N, no correction is applied: a size inferred from the seats seen so far is itself a lower bound, so counting every known seat is not a census. The range covers students who weren't counted, not classifier mistakes.

Seat counting. Before step 3, every box that persisted about 1.5 s became a permanent seat. On the 6-minute study-hall clip the inferred class grew to 78 for about

50 students; with the rule it is 52, and 31 on the exam clip with about 30 students.

Checks. On the Pi, over 2,060 frames per setting, the shown estimate was never outside its range and the range was never zero-width. Without `--class-size` the ranges are wide, 25-37 points for 20-30 counted students, which is honest for samples that small.

4.4 Per-student decision: rules versus learned models

The learned classifiers beat the rules on SCB by a wide margin, but on real classes they call attentive students distracted, so the rules are the system's default and the learned models are an opt-in comparison.

The two approaches. The rules combine pose cues with a noisy-OR: head yaw beyond the seat's calibrated baseline, head dropped below it, face hidden while the shoulders are visible, torso turned, a hand at the face; a raised hand always counts as focused. Each seat learns its own attentive pose in its first 120 s of observation, and labels are smoothed (EMA with hysteresis). The learned models (gradient-boosted trees, logistic regression, an MLP 64-32) take the keypoints and the same cues as features and are trained on SCB students with the deployed INT8 m model's poses, with and without tiles. They run in NumPy on the Pi.

On SCB (58,185 front-view students, 182 lessons, 5-fold cross-validation grouped by lesson):

Table 4.4 Per-student decisions on SCB (lesson-grouped 5-fold cross-validation)

Decision	AUC	Within-image AUC	Balanced accuracy
Rules (single image, uncalibrated)	0.55	0.54	56%
Logistic regression	0.772	0.735	71.3%
Gradient-boosted trees	0.855	0.840	77.0%
MLP 64-32	0.863	0.838	78.0%

The within-image AUC is the figure to quote: most SCB images contain only one class, so a model can partly recognise the image instead of the pose, and comparing students within the same image removes that shortcut. The rules are near chance here partly by design: they compare each student with their own calibrated pose, which a single image cannot provide.

On real classes the ranking reverses:

In the lecture clip, students sitting upright and looking at the board were labelled distracted, and students reading with their heads down focused (median head height above the shoulders 0.45 versus 0.32). The reason is a gap in SCB's labels: its focused classes are read, write and hand-raising, all head-down or hand-up, while its distracted classes include turning the head and discussing, both head-up. SCB

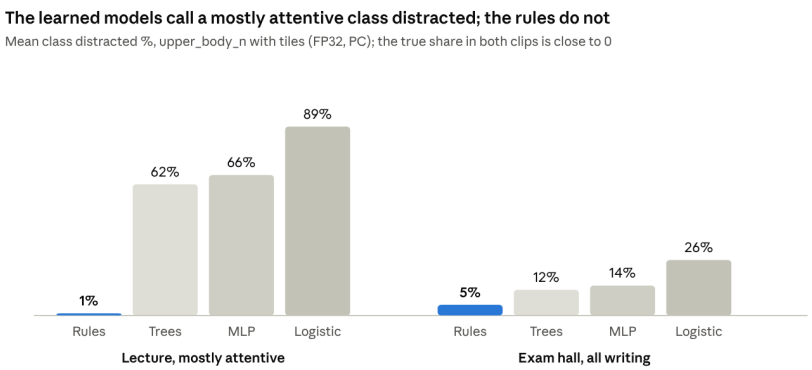


Figure 4.3 Mean class distracted share on the lecture and exam-hall clips: rules against learned classifiers (upper_body_n with tiles, FP32)

has no "listening" or "looking at the board" label, the most common attentive state in a lecture, so a model trained on it learns that a raised head means distracted. Removing the head-height features did not fix it in a test, and the learned models also depend strongly on pose quality: with n and no tiles, trees gave 36% on the exam clip on the Pi, against 9-13% with m.

On the deployed INT8 models the picture is the same: in the lecture clip with m and tiles, the rules gave 0.0% and the trees 48%, logistic regression 74% and the MLP 52%.

Against human labels. 301 student-moments from the lecture and exam clips were labelled blind (focused, distracted or can't tell), sampled so that each disagreement between the rules and the trees was represented. Only 13 of the 275 judgeable moments (4.7%) were distracted, 4.2% when re-weighted to the whole clips. Pooled over both clips:

Table 4.5 Per-student decisions against 301 blind human labels (lecture and exam-hall clips)

Decision	False alarms on focused moments	Specificity (95% CI)	Distracted moments caught
Rules	27 / 262	90% (85-93%)	5 / 13
Gradient-boosted trees	114 / 262	56% (50-62%)	3 / 13
MLP 64-32	120 / 262	54% (48-60%)	2 / 13
Logistic regression	199 / 262	24% (19-30%)	5 / 13

Re-weighted to the whole clips, the rules flag 1% of lecture moments and 3% of exam moments, close to the 4% truth, while the learned models flag 56-94% of lecture moments. Thirteen distracted moments are too few to estimate recall; the staged sessions measure it.

Held-out class (clip_5). The deployment setting kept its low false-alarm rate on a class recorded after every threshold was fixed. 150 moments were labelled blind: 121 focused, 28 distracted and 1 can't tell, or 6.8% distracted re-weighted to the

whole clip. The moments were sampled from m-with-tiles detections, and a moment a run did not detect counts as not flagged. No frames of clip_5 are shown, because its students are minors.

Table 4.6 Per-student decisions against 150 blind human labels on the held-out class (clip_5)

Model, tiles on	Decision	False alarms on focused	Specificity (95% CI)	Distracted caught	Flagged share of the clip (re-weighted)
n	Rules	11 / 121	91% (84-95%)	8 / 28	4.7%
n	Trees	35 / 121	71% (62-78%)	3 / 28	17.7%
n	Logistic	44 / 121	64% (55-72%)	11 / 28	33.9%
n	MLP	42 / 121	65% (56-73%)	9 / 28	28.4%
m	Rules	43 / 121	64% (56-72%)	21 / 28	10.1%
m	Trees	51 / 121	58% (49-66%)	12 / 28	11.4%
m	Logistic	55 / 121	55% (46-63%)	10 / 28	35.4%
m	MLP	37 / 121	69% (61-77%)	9 / 28	20.0%

- **Rules on n:** 91% specificity, against 90% on the first label set. They flag 4.7% of the clip against 6.8% labelled, and catch 8 of 28 distracted moments.
- **Learned models:** they again false-alarm far more, at 55-71% specificity.
- **Rules on m:** they catch the most (21 of 28) but flag 36% of focused moments. Six of those 43 false alarms are on one front-row student.
  - The camera sees under that student's desk, so the box switches between upper body and full body. The models learned full-body boxes from COCO, whose person boxes cover the legs.
  - The `body_turned` cue divides shoulder width by box height, so the tall box reads as a turned torso.
- **Without tiles:** n leaves 92 of the 149 moments undetected.
- **Seat counting:** with tiles, the analyzer found 13-17 seats for 10 students.
  - Seats are anchored at the top of the box. At this close range, a front-row head bending to write moves further than the seat-matching radius and opens a new seat.
  - The full-body box adds a second seat on the same student.
  - So after 2:30 the class indicator averages 11-15% for the rules with tiles, against 6.8% labelled.

Staged two-person session (clip_4). The author and a colleague, both adults, sat at one table filmed by a phone mounted high at the front, and followed a script for 4:45. Recording a script gives the true state of each person every second. After 55 s of attention, the script has seven distracted events:

- looking to the side (twice)
- a phone under the table and a phone on the table
- talking to each other
- turning around

There is also a minute of writing, which should stay focused. Results on the INT8 m model (the n model in brackets where it differs):

Table 4.7 Staged two-person session (clip_4), INT8 m model

Decision	Distracted seconds caught	Precision	False alarms, attentive / writing	Events missed
Rules, original head-turn cue	11%	100%	0% / 0%	6 of 7
Gradient-boosted trees	40% (17%)	100%	0% / 0%	2 of 7
MLP 64-32	40% (11%)	100%	0% / 0%	3 of 7
Logistic regression	34% (19%)	64%	44% / 0%	3 of 7
Rules with the head-offset measure (new default)	56% (45%)	100%	0% / 0%	2 of 7

No decision caught the phone, whether on the table or under it. Head, wrists and elbows sit within about 0.2 shoulder-widths of where they sit when writing, so 13 upper-body keypoints cannot separate the two. The learned models' figures predate a seat-matching fix described below, so their turn-around seconds are slightly undercounted.

Why the rules missed head turns. The original head-turn cue measures yaw as the nose's position between the two ears. In profile the far ear disappears (confidence about 0.01), and the code falls back to estimating the head centre from the remaining ear. That ear has moved with the head, so the estimate cancels most of the turn:

The nose's offset from the shoulder centre, in shoulder-width units, shows every turn: about ±0.05 when attentive and ±0.35-0.54 when turned. It was added as a second head-turn measure, compared with each seat's own calibrated offset. The cue takes whichever of the two measures shows the larger turn. Like yaw, each seat's offset baseline is kept within 0.15 of the room's, so a student who looks away during calibration does not learn "turned" as attentive. The same session also showed that a student who turns around shrinks to a head-and-shoulders box whose centre jumps

Table 4.8 Ear-based yaw and nose offset at attentive and turned moments of clip_4

Moment	**Person**	**Ear-based yaw**	**Nose offset from the shoulder centre**
Attentive	blue / grey	−0.15 / +0.07	−0.07 / +0.05
Talking to each other	grey	+0.03	−0.35
Looking to the side	blue	−0.31 (0.16 from his baseline)	−0.54

by about a third of its height, so seats are now anchored at the top of the box.

Validation of the fix. The offset's thresholds were chosen with clip_4, so its clip_4 recall is optimistic. Its cost was therefore measured on everything else, in a sweep over 16 settings on the saved INT8 skeletons. At the chosen ramp (0.2-0.4, other thresholds unchanged):

Table 4.9 Effect of the head-offset measure across all evidence

	n, before → after	**m, before → after**
clip_4 distracted seconds caught	11% → 45%	11% → 56%
Human labels: false alarms on focused moments	4% → 5%	6% → 7%
Lecture class % (truth near 0)	4.7% → 4.7%	0.0% → 0.0%
Exam-hall class %	3.6% → 8.4%	11.7% → 14.6%

The only cost is 3-5 points on the exam clip. Some of that may be real, since the human labels found exam students looking around. Loosening the yaw thresholds alone reached only 25% on clip_4. The measure is now the default, and the 9-person session is kept as the held-out test.

The measure helps the rules only. The learned classifiers decide from the features they were trained on, which do not include it. It also cannot be learned from SCB: it needs a per-student baseline, and on SCB's single images with small faces it does not separate turned heads from reading (AUC 0.53).

In profile the far ear's confidence falls to about 0.01, so the ear yaw reads near zero while the nose sits well to one side of the shoulders. The head-offset measure picks up these turns.

The benchmark accuracy of a classifier trained on a public action dataset therefore does not transfer to measuring attention; per-student calibration does better. Making a learned model usable would need training labels that include attentive, head-up students from the target classrooms, which the pending human labels and staged session begin to provide.

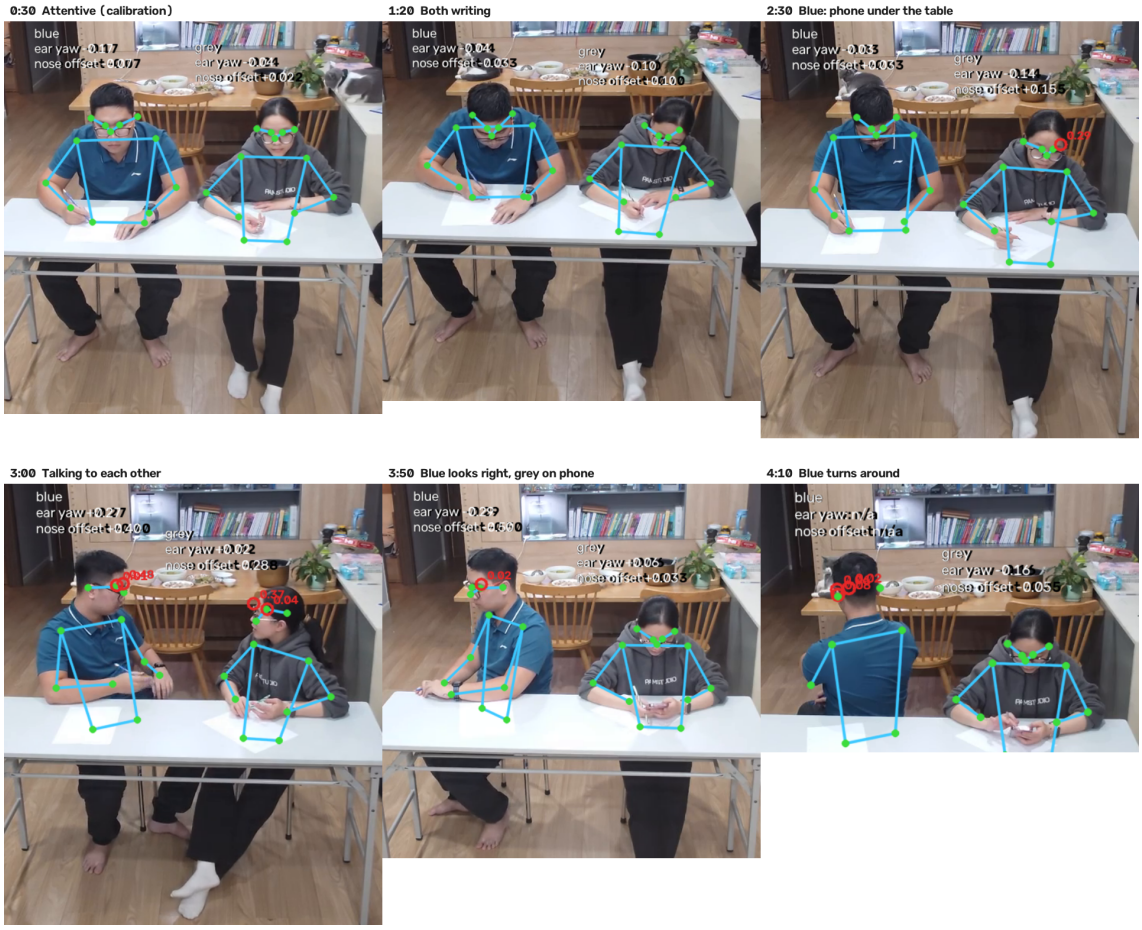


Figure 4.4 what the pose model sees on clip_4, with the two head-turn measures per person

4.5 Edge performance

In the deployment setting, n with tiles in real time, the system ran for 15 minutes at 30 FPS with 45 ms latency, 4.3 W measured on the Pi board plus 2.5 W for the Hailo-8 (vendor figure), a peak SoC temperature of 60 °C and no throttling. The pipeline is bound by the Hailo-8, not the CPU. All figures below were measured on the Pi 5 with official Raspberry Pi active cooling on its fan header, on 10 November 2026.

Hailo-8 alone (`hailortcli benchmark`, INT8 HEFs):

Table 4.10 Hailo-8 throughput and latency per model (hailortcli benchmark)

Model	FPS	Hardware latency (ms)
n	131.5	5.7
s	69.5	11.9
m	31.5	26.7

End to end (lecture clip, 1280×720, GStreamer engine, engagement on, as fast as possible):

Table 4.11 End-to-end performance on the Raspberry Pi 5 (lecture clip, 1280×720)

Model	Tiles	FPS	CPU, whole Pi	Memory of hcpose (MB)	Max SoC temp (°C)	Pi board power (W)
n	off	119.1	79%	332	61.1	5.42
n	on	32.8	46%	286	60.6	4.84
s	off	69.3	47%	329	60.1	4.77
s	on	17.4	26%	294	59.5	3.87
m	off	31.5	23%	339	58.4	3.19
m	on	7.9	14%	307	57.3	2.82

End-to-end FPS matches the Hailo-only figures, and four passes per frame divide them by exactly four. The CPU's main cost is the engagement analysis, about 2.8 cores at 119 FPS. Board power is lower for the slower settings because the Pi waits on the accelerator.

Latency. Frames were released at the clip's 25 FPS, as a camera releases them, and timed from that point to the sink with GstShark's interlatency tracer [20]: median 20 ms for n (95th percentile 23), 45 ms for n with tiles (53), and 40 ms for m (47). m with tiles cannot keep up with a 25 FPS camera (7.9 FPS): frames queue and latency settles at 2.5 s, so a live m-with-tiles deployment needs a leaky queue or a lower camera rate.

Thermal behaviour over two 15-minute runs:

Idle, the Pi drew 2.33 W at 50 °C.

Power. Board power is the sum of voltage × current over the Pi 5's PMIC rails (`vcgencmd pmic_read_adc`). It excludes the AI HAT and USB devices, so the

Table 4.12 Thermal behaviour over 15-minute runs

Run	FPS	SoC temperature after 5 / 10 / 15 min (°C)	Max (°C)	Throttling	Pi board power (W)
n with tiles, real time (deployment)	30.0	56.8 / 57.9 / 56.8	60.1	none	4.26
m with tiles, as fast as possible	7.9	53.5 / 55.6 / 54.6	56.2	none	2.70

Hailo-8 is added from vendor figures:

Table 4.13 Vendor power figures used to complete the measured board power

Part	Figure	Source
Raspberry Pi 5, typical bare-board active current	800 mA at 5 V ≈ 4 W	Raspberry Pi documentation [21]
Raspberry Pi 5, recommended supply	5 V, 5 A (27 W)	same
Hailo-8, typical power	2.5 W	Hailo [5]
Hailo-8 M.2 module, maximum	8.65 W	Core Electronics listing [22]
AI HAT+, operating ambient	0-50 °C	AI HAT+ product brief [23]

The deployment setting therefore draws about 4.3 W (measured) + 2.5 W (typical, vendor) ≈ 6.8 W, well within one 27 W USB-C supply. The measured board figure agrees with the vendor's 4 W typical. 26 TOPS at 2.5 W is about 10 TOPS/W (own calculation from Hailo's figures).

4.6 Privacy by design

The system never stores or transmits an image by default: each frame lives in memory for one inference, and the only default output is a class-level share shown on the teacher's screen.

- **Frames** are decoded into RAM, passed to the Hailo-8 and freed. Nothing is written to disk unless `--output` is set, which is meant for development and evaluation, not for classes.
- **Skeletons** (13 keypoints and a box per person) carry no pixels. They identify nobody by themselves, but tied to a seat and a time in a known class they could point to a student, so they are treated as personal data and kept in memory.
- **Seat states** use anonymous seat numbers that exist only for one session. There is no face recognition, no name list and no link between sessions, except an optional calibration file (`--calibration`) that stores each seat's attentive pose

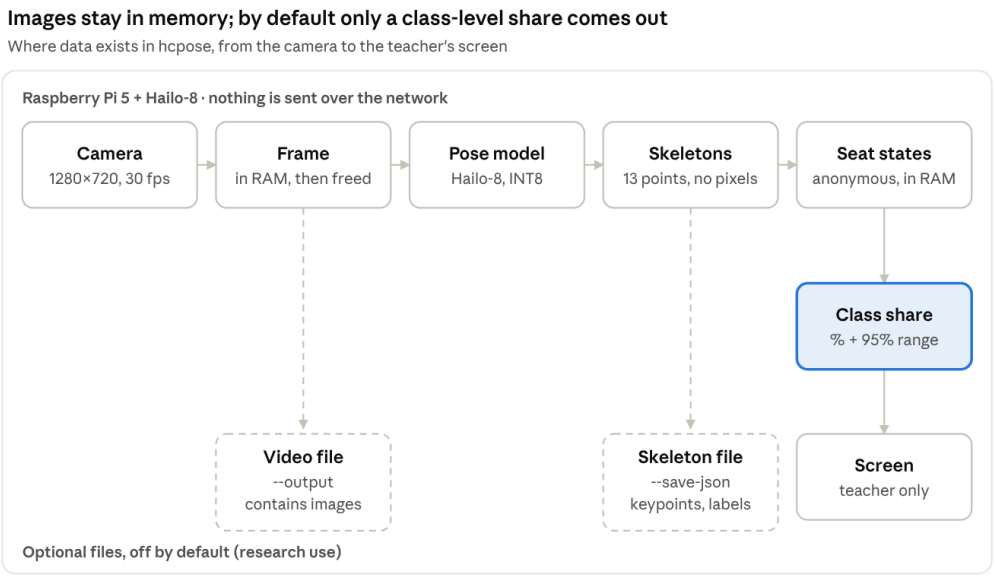


Figure 4.5 Data flow in hcpose with default settings; optional outputs are dashed

and position, not identities.

- **The class share** with its range is the only number meant to leave the program. It describes the class, not a student, and is not personal data.
- **The screen** shows the camera image with labels during development. For classroom use, `--display class` limits the overlay to the class number, and `--headless` shows no window at all.
- **Network:** hcpose opens no network connection; the device can run fully offline.

What a saved record holds, when `--save-json` is used: one line per frame, from clip_4 at 3:50, shortened to one person and three of the 13 keypoints. There are coordinates, scores and states, but no pixels and no identity. The seat number exists only within the session.

```
{"type": "frame", "t": 229.97, "w": 1280, "h": 720,
 "class": {"visible": 2, "counted": 2, "focused": 1, "distracted": 1,
           "class_pct_distracted": 33.3, "class_ci": [6.1, 79.2],
           ↪  "class_size": 3},
 "persons": [{"score": 0.75, "box": [663.9, 296.1, 863.6, 709.2],
              "keypoints": [[771.9, 356.1, 0.98], [776.5, 348.7, 0.91],
              ↪  [761.6, 349.6, 0.95], "…"],
              "engagement": {"seat": 2, "state": "focused", "score":
              ↪  0.0, "reason": null, "calibrated": true}}]}
```

Compared with a cloud service, nothing has to be uploaded, stored by a third party or transferred across borders, which removes the cross-border transfer obligations and most of the breach risk.

4.7 Legal fit in Vietnam

Since 1 January 2026, personal data in Vietnam is governed by the Law on Personal Data Protection (No. 91/2025/QH15) [12] and its implementing Decree 356/2025/ND-CP, which replaced Decree 13/2023/ND-CP. Keeping video on the device and reporting a class-level share lowers, but does not remove, the obligations of a school using the system. This section is an analysis for the thesis, not legal advice.

Table 4.14 Provisions of Law 91/2025/QH15 and Decree 356/2025/ND-CP and their consequences for the system

Provision	**What it says**	**Consequence for the system**
Personal data (Law, Art. 2.1)	data that "identifies or helps identify a specific person"	Camera frames are personal data. Keypoints tied to a seat and a time in a known class can help identify a student, so per-student skeletons and labels are treated as personal data too; the class-level share is not.
Sensitive data (Law, Art. 2.3; Decree 356, Art. 4)	data tied to privacy, on a Government list that includes biometric data	hcpose does no face recognition and keeps no identity. Whether body keypoints for behaviour analysis count as biometric is not settled, so the design treats per-student data as if it were sensitive: kept in memory by default, never transmitted.
Purpose limitation (Law, Art. 3.2)	processing "for specific and clear purposes"	One stated purpose: an aggregate attention indicator for the teacher. No per-student report, ranking or grading.
Consent (Law; Decree 356, Art. 9)	consent per purpose; no default or bundled consent	The school informs students and parents and obtains consent for the camera and this purpose before deployment.
Children (Law, Art. 24.2)	legal representatives act for children; children aged 7 or older must also consent where their private life is disclosed	Most pupils are minors: parental consent is needed, and older pupils should be informed and asked as well.

(continued)

Provision	What it says	Consequence for the system
Recording in public places (Law, Art. 32)	allowed without consent only for listed public activities, with notice	A classroom lesson is not one of the listed cases, so the consent route above applies.
Impact assessment (Law; Decree 356)	a processing impact assessment, updated when processing changes	A school deploying it prepares a DPIA. The small-business grace period does not cover processing sensitive data directly.
Breach notification	within 72 hours	With nothing stored by default there is little to breach; any saved files fall under it.

What the design contributes. Data minimisation and storage limitation are built in: frames exist only in RAM for the time of one inference, nothing leaves the device, and the default output is a class-level number. What stays the school's job: notice, consent, a DPIA, and keeping the optional research outputs (annotated video, skeleton files) switched off in live use.

Sources: [12, 24–27]. Article numbers follow the English translation; check them against the official Vietnamese text before citing.

4.8 Limitations and threats to validity

The largest gap is ground truth: no result so far measures per-student accuracy on a real lecture, only on SCB actions or through the context of mostly attentive clips.

- **Proxy labels.** SCB labels actions (reading, turning, discussing), not attention, and comes mostly from Chinese K-12 classrooms. About 2,000 SCB students carried both a focused and a distracted label across duplicate images, mostly "bowed head" versus "reading or writing", so even the annotators could not separate these.
- **Clip evidence is contextual.** The three clips have no labels; "the true share is close to 0" rests on what the class is doing. They show false alarms, not missed distraction. The human labels and the staged session address this.
- **Few rooms, third-party footage.** Three third-party clips from three rooms, one of them with a camera outside the intended mounting, plus one small class recorded for this work. Results may not carry over to other layouts, lighting or age groups.
- **Camera height.** With a camera at desk height, a head bowed over writing

hides the face and reads as "turned away": the rules gave 44% distracted in a study hall where everyone wrote. The design assumes a camera mounted high at the front; a lower `face_hidden` weight (0.3) halves the error for low cameras.

- **Calibration time.** Each seat needs 120 s of usable observation. In a 6-minute dense clip only 20 of 52 seats finished calibrating, so short sessions run mostly on room-level baselines.
- **Small classes, wide ranges.** With 20-30 counted students the 95% range spans 25-37 points unless the head count is given. This is a property of the sample size, not a defect, but it limits how finely the indicator can be read.
- **Hand-set rule weights.** The rule weights were set by design, not fitted, and were deliberately not tuned on the clips used to evaluate them; one documented change, the low-camera weight, is an option rather than the default.
- **Mixed precision of the evidence.** The two charts and the human-label scores use the n model in FP32 on a PC; the INT8 runs on the Pi reproduce the class-level numbers within a few points. m with tiles also cannot keep up with a live 25-30 FPS camera without dropping frames.
- **Staged session.** Adult volunteers acting scripted behaviour in a small room are easier to see and more exaggerated than real distraction, so its recall is an upper bound.
- **Close cameras and full-body boxes.** In a small room filmed from close range, seats multiply (13-17 for 10 students in clip_5), and a full-body box on a student whose legs are visible reads as a turned torso. Both come from geometry tied to the box: the seat anchor at its top and the box height in `body_turned`. Keypoint-based anchors and scales would avoid them; they were not changed after clip_5, to keep it a held-out test.

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 Answers to the research questions

RQ1, feasibility at the edge: yes. On a Raspberry Pi 5 with a Hailo-8, the deployment setting (n model with tiles) ran 15 minutes in real time:

- 30 FPS with 45 ms latency
- 4.3 W measured on the board, plus about 2.5 W for the accelerator
- peak SoC temperature 60 °C, no throttling

The pipeline is limited by the accelerator, not the CPU. The m model with tiles reaches only 7.9 FPS, enough for analysis but not for a live camera without dropping frames.

RQ2, coverage: substantially improvable without retraining. Tiling the back of the room raised the share of SCB students found from 49% to 76%, and in the far third from 17% to 74%. It doubled the students found in a dense study hall. Removing same-skeleton duplicates and anchoring seats at the head made the inferred class size match the real one (52 for about 50 students, 31 for about 30).

In a small room filmed at close range (clip_5), seats still multiplied, to 13-17 for 10 students, so the class-size estimate holds for the camera distances it was developed on, not yet for very close cameras.

RQ3, the per-student decision: calibration beats benchmark accuracy. Classifiers trained on SCB rank SCB actions well (within-image AUC 0.84) but transfer badly:

- **Lecture clip:** they call 48-89% of an attentive lecture distracted.
- **Human labels:** they wrongly flag 114-199 of 262 focused moments.
- **Cause:** SCB has no "listening" label, so they learn that a raised head means distracted.

The calibrated rules have very few false alarms (specificity 90% on human labels; 0-5% on the lecture). They first missed most real distraction (11% on the staged session), because the ear-based head yaw collapses in profile. A second, body-relative head-turn measure raised that to 45-56%, with no new false alarms there and a small cost on real classes. On a 10-student class labelled after all thresholds were fixed (clip_5), the rules on the deployment setting kept 91% specificity. Phone use remains invisible to pose.

RQ4, the class indicator: yes, with honest width. Seat memory, zone weighting and a Wilson interval around the shown value give a share that always lies inside its range. Without the real head count the range is wide (25-37 points for 20-30 students), which reflects how little a sample that small can say. Giving `--class-size` narrows it.

5.2 Future work

1. **A held-out test at classroom scale.** Run the 9-person scripted session (research/staged/script_v2.csv). clip_5 confirmed the rules' specificity on a class nothing was tuned on, but its recall still rests on one two-person session.
2. **Classroom-specific training labels.** The labelling and staged-session tools can grow a dataset that includes attentive, head-up students, the case SCB lacks. A learned decision trained on it, with per-seat baselines as features, could combine the classifiers' sensitivity with the rules' specificity.
3. **Phone use.** Detecting phones needs an object detector. On the Hailo-8 it could run on the same frames, at a cost in frame rate. Only a per-seat "phone visible" flag would leave the device.
4. **Longer field deployments.** Weeks of use would show drift in calibration, the effect of lighting and seating changes, and whether teachers find the indicator useful. That needs a study with consent, designed with the school.
5. **Live m with tiles.** A leaky queue or a lower camera rate would let the most accurate setting run live without growing latency.
6. **Keypoint-based geometry.** Anchor seats on the shoulders and measure torso rotation against the visible upper body instead of the box. That removes the two failures seen at close range in clip_5: seats multiplying and a full-body box reading as a turned torso.
7. **Teacher-facing evaluation.** The real test of a class-level indicator is whether it helps teaching, which needs a user study rather than more accuracy metrics.

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